

Spyridon Liaskonis

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EDUCATION

2020 - Present Integrated Masters Degree (5 years) in ELECTRICAL AND COMPUTER ENGINEERING
University of Thessaly, Volos, Greece

RESEARCH INTERESTS

Hardware Accelerators, Embedded Systems, Deep Learning, Computer Architecture, Machine Learning, High-Performance Computing.

EXPERIENCE

FPGA / Embedded Systems Engineering Intern - TIMA Laboratory, Univ. Grenoble Alpes February 2026 - Present

Multi-FPGA acceleration of neural network inference on Xilinx VC709 (Virtex-7) platforms.

- Designed and brought up an **Aurora 64B/66B** high-speed serial link over SFP+ between accelerator cores
- Extended the `nnawaq` VHDL accelerator generator to chain multiple accelerator cores across an FPGA cluster, and adapted the host-side **C++ inference driver** (PCIe/DMA, AXI) and the **Vivado Tcl** build flow to the new cascaded architecture.
- Developed a bandwidth- and resource-aware min-max **dynamic-programming partitioner** that automatically splits streaming DNN accelerators across a linear multi-FPGA cluster, treating the inter-chip link rate as a constraint of the optimisation.
- Evaluated end-to-end flow on ResNet-8/18/50 and MobileNet-V1/V2 at 250 MHz over single-lane Aurora (Vivado 2025.1).

Technologies/Tools: C, C++, Verilog, VHDL, Tcl, Python, Vivado, Aurora 64B/66B, SFP+, PCIe, AXI/AXI-Stream, I²C, UART, git

Teaching Assistant - University of Thessaly

September 2024 - February 2026

- ECE219 Computer Organization and Design** (Sep 2024 - Feb 2026): supervised and examined laboratory assignments on MIPS assembly, Verilog implementation of a MIPS CPU, and performance analysis of compute-intensive kernels on x86.
Reference: [Prof. Nikolaos Bellas](#)
- ECE220 Numerical Analysis** (Feb 2025 - Jun 2025): ran lab sessions on numerical methods and their implementation in MATLAB; supported students individually through assignments.
Reference: [Prof. Panagiota Tsompanopoulou](#)

SKILLS

- Embedded & low-level:** C, C++, bare-metal ARM, embedded Linux (PetaLinux), I²C, UART, PCIe, AXI / AXI-Stream, DMA, JTAG, device bring-up and debugging
- HDL & FPGA:** Verilog, VHDL, HLS C/C++, Vivado, Vitis, Vitis HLS, Vivado Tcl scripting, timing and resource analysis, AMD/Xilinx Virtex-7 and Zynq-7000/UltraScale+ MPSoC
- Software & tooling:** Python (NumPy, Matplotlib, pandas), MATLAB, CUDA, OpenMP, MPI, git, make, CMake, Bash, Linux (daily driver + administration), LaTeX
- ML frameworks:** TensorFlow/Keras, PyTorch, QKeras, hls4ml, FINN, Brevitas, ONNX

THESIS

Automatic Quantization and Pruning of Neural Networks for Edge Deployment with hls4ml (Ongoing, defence expected September 2026)

Developing **rl4hls**, a framework that automates optimisation and deployment of neural networks onto resource-constrained FPGAs. Built on **hls4ml**, the open-source HLS toolchain developed for low-latency inference in **high-energy physics trigger systems**.

- Built a Quantizer that parses an arbitrary Keras model and emits an equivalent quantized QKeras implementation, plus an end-to-end pipeline from Keras model to synthesised FPGA design.
- Trained reinforcement-learning agents (A2C, PPO, SAC) to search the mixed-precision quantization space.
- Built an analytical estimator for DSP/II of hls4ml networks, as well as calibrated BRAM, FF, LUT estimators to eliminate redundant HLS synthesis runs during the search.
- Evaluated on high-energy-physics jet classification and anomaly-detection benchmarks.

SELECTED ACADEMIC COURSES - PROJECTS

ECE 340: Embedded Systems (Grade 9.75/10)

Technologies: C, OpenCL, Verilog, Vivado, Vitis HLS, PetaLinux

- Hardware acceleration of the Smith-Waterman Local Sequence Alignment algorithm for Genomics using the Vitis High-Level Synthesis toolset, on a low-power FPGA MPSoC.
- Performance measurement of bare-metal embedded software on an ARM processor.
- IEEE 754 compatible Floating-Point Adder in Verilog on an FPGA.

[Project Repository](#)

ECE 415: High-Performance Computing (Grade: 9.5/10)

Technologies: C, CUDA, OpenMP, AVX

- Parallel implementation of a Sobel Filter with OpenMP.
- GPU acceleration of a separable convolution kernel with CUDA.
- GPU acceleration of a histogram equalization algorithm with CUDA.
- CPU/GPU acceleration of an N-body simulation with OpenMP/CUDA.

[Project Repository](#)

ECE 338: Parallel Computer Architecture (Grade 9/10) - Based on Computer Architecture: A Quantitative Approach by Patterson & Hennessy

Technologies: Intel Pin, ChampSim

- Implementing machine learning-based branch prediction for hard-to-predict branches.
- Developing a neural network trained specifically on hard-to-predict (H2P) branches, inspired by the works of [Branch-Net](#) and [Tarsa et al.](#)
- Testing and evaluating the network on SPEC 2017 benchmarks using the [ChampSim](#) simulator.

ECE 461: Machine Learning for Data Science and Analytics (Grade: 10/10)

Technologies: Python, Pandas, Jupyter, Hugging Face, TensorFlow, PyTorch, Transformers, LaTeX

- Implemented probabilistic machine learning models, including regression, neural networks, decision trees, SVMs, and ensemble methods.
- Applied optimization, probability, and statistical methods to model training and evaluation.
- Designed and presented large-scale ML projects in Jupyter with formal LaTeX reporting, including deep learning for sentiment analysis and time-series forecasting for weather prediction.
- Explored generative pre-trained transformer (GPT) models and transfer learning workflows.

ECE 333: Digital Systems Lab

Technologies: Verilog, Xilinx Vivado

- A 7-Segment Display Driver to display a scrolling message.
- A UART serial communication system.
- A VGA driver used to drive a conventional monitor and depict a variety of images.

[Project Repository](#)

ECE 445: Parallel & Distributed Computing (Grade 9/10)

Technologies: C, OpenMP, OpenMPI

- Designed and implemented parallel algorithms for matrix multiplication and the Jacobi method using OpenMP, focusing on thread distribution, scheduling, and performance optimization.
- Designed and implemented parallel algorithms for calculating communication costs in a distributed system, using MPI to measure point-to-point and broadcast communication times, and optimizing performance for matrix-vector multiplication and sorting algorithms.

[Project Repository](#)

ECE 513: Circuit Simulation Algorithms (Grade 10/10)

Technologies: C++, FLEX, Bison, Eigen, CMAKE

- Developed a complete circuit simulation program like SPICE in C++.
- Implemented parsing, equation formulation, and solution techniques (direct and iterative methods) for linear circuits.
- Utilized sparse matrix techniques and external libraries (Eigen) for efficient computation.
- Performed transient analysis to simulate circuit behavior over time.

[Project Repository](#)

LANGUAGES

English: Professional Working Proficiency (ECCE Certification)

Greek: Native Speaker